

Sanay Tralshawala

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EDUCATION

Carnegie Mellon University (CMU), Pittsburgh, Pennsylvania

Dec 2026

Master of Science in Artificial Intelligence Engineering-Mechanical Engineering | Concentration: Robotics & Controls | GPA: 4.00/4.00

Relevant Coursework: Machine Learning in Production, Multimodal Machine Learning, Modern Control Theory, Computer Vision

Rensselaer Polytechnic Institute (RPI), Troy, New York

May 2025

Bachelor of Science in Mechanical Engineering | Tau Beta Pi Scholarship | Class of 1957 Spectrum Award | GPA: 3.95/4.00

SKILLS

AI/ML– Agentic AI, LangChain, Model Context Protocol (MCP), Retrieval-Augmented Generation (RAG), Machine Learning, Computer Vision, AI Safety, Embodied AI Safety

Engineering – MATLAB/Simulink (Control Systems Toolbox, Optimization Toolbox), Control Theory, Robot Kinematics, Numerical Optimization, Probabilistic Risk Assessment (PRA), Finite Element Analysis (FEA)

Systems, Pipelines & Tooling – Docker, Kafka Streaming, Scalable Pipelines (PySpark), Google Cloud, PGAdmin

Programming – Python (PyTorch, TensorFlow, scikit-learn, SciPy, Pandas, OpenCV, JAX), C++, SQL, R, Java, Cypher (Neo4j)

EXPERIENCE

Graduate Researcher - Computational Engineering and Robotics Lab (CERLAB), CMU

Oct 2025 - Present

- Trained a vision foundation model backbone (triplet loss + cross-entropy) for object re-identification in a SLAM pipeline, converging to ~1.2 loss with ~0.80 mAP on the best configuration.

Product Development Model-Based Engineering Intern - The Boeing Company

May 2026 - Aug 2026

- Piloted Model Context Protocol (MCP) integrations across 6 open-source tool workflows and connected an AI copilot to an internal design tool, applying software development best practices to demonstrate reusable, production-style agentic tooling.
- Demonstrated a LLM-guided optimization method for data-chunking over a disruptive network with Docker network.
- Engineered a document-pairing framework, intended to match 5,000 variable-length documents against 1,500 abstract snippets using K nearest neighbor, cosine-similarity, thresholding and LLM-based document comparisons.

Semantic Segmentation of Satellite Imagery - Deep Learning Engineering

Apr 2026 - May 2026

- Trained DINOv3 ConvNeXt-Tiny on FLAIR-1 aerial imagery, achieving 0.491 mean Intersection over Union (mIoU), a 6.7% improvement over a ResNet34 U-Net baseline across six land-cover classes.
- Ran conformal prediction analysis across six model configurations and surfaced severe minority-class under-coverage that high average confidence scores had been masking.

Trajectory Prediction as Forward Reachability for Airport Surface Safety - Embodied AI Safety

Jan 2026 - May 2026

- Calibrated forward reachable sets for airport surface operations using conformal prediction and data-driven trajectory forecasts to flag anomalous aircraft behavior ahead of conflict.
- Applied aircraft-specific kinematic feasibility weighting to constrain reachable envelopes, outperforming kinematic-only and unicycle baselines on a documented incursion sequence.

Agentic Bus Routing from Disruption Data - AI Agents for Engineers

Jan 2026 - May 2026

- Designed a 2-stage LangChain news-parser agent with confidence-based filtering to extract article fields.
- Owned architecture and delivery of a 6-stage, multi-agent LLM pipeline spanning 4 core disruption/event categories, integrating with Google Maps APIs end-to-end from data ingestion to actionable stop-substitution plans.

Hardware Digital Intern - Northrop Grumman Corporation, Microelectronics Center

Jun 2025 - Aug 2025

- Designed 11 custom mechanical parts and assemblies for a test fixture, utilizing Commercial Off The Shelf components when possible, performed structural analysis on test equipment using Solidworks.

Risk and Reliability Analyst (Student Co-Op) - United States Nuclear Regulatory Commission

Jun 2024 - May 2025

- Ran component and system failure analyses across 8 major nuclear plant, evaluating about 4 failure scenarios per plant using Probabilistic Risk Assessment (PRA) models. Assisted in validating proposed regulations in technical specifications.

Legislative Intern - Office of Congressman Paul D. Tonko, U.S. House of Representatives

May 2023 - Jun 2023

- Researched and responded directly to constituent phone calls and correspondence on behalf of the Congressman with a high volume of live conversations on time-sensitive and often contentious topics.

LEADERSHIP

Senior Engineering Ambassador (EA) - RPI Engineering Ambassadors

Mar 2023 - May 2025

- Developed 3 hands-on engineering learning modules, including a semiconductor module simulating the photolithography process using nail polish, and delivered them across 12+ school visits over two years, reaching 800+ students.

Director of Sponsorship - HackRPI Organizing Team

Jan 2023 - Jan 2024

- Obtained, communicated, and facilitated conversations with 30+ potential sponsors and raised \$100,000+ in monetary and prize sponsorship; increased participation from 200 to 500+ with a more diverse attendance for the annual event.